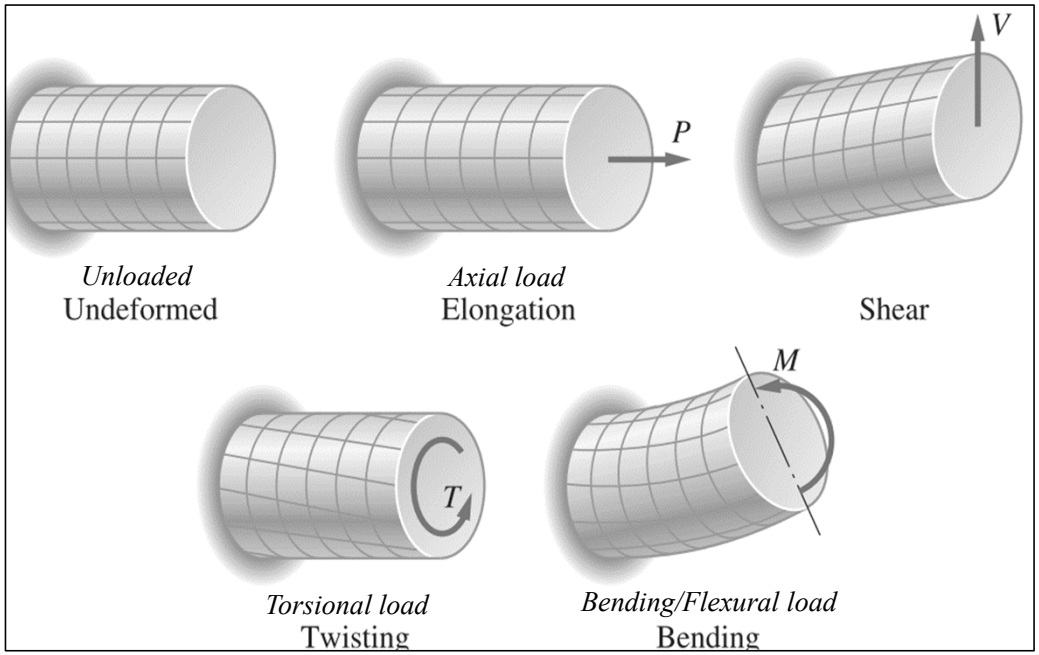
Stress and Strain under Axial Loads

Recall;

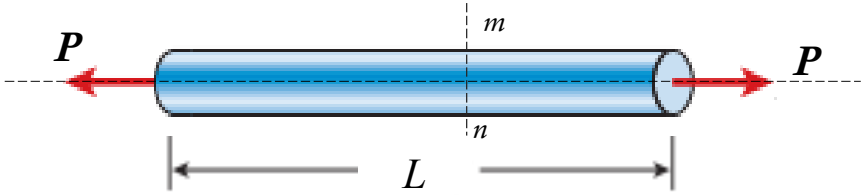
- Stress is a quantity that accounts for the intensity and distribution of internal reactions (forces and moments).
- Strain is the quantity measures the intensity of the deformation in a stressed body.
- External loads are also of several types;
  - Axial loads
  - Torsional loads
  - Bending loads
- Stresses depend on the type of external load;
  - Tensile stress and compressive stress
  - Bending/Flexural stress
  - Shear Stress
  - Thermal stress



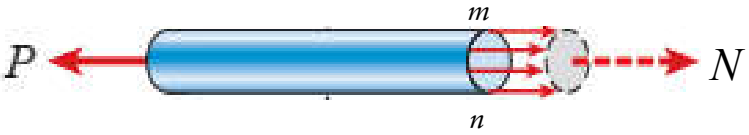
Source: Mechanics of Materials, Pytel and Kiusalaas

Axial loads; Normal Stress and Strain

- Axial loads are applied along the longitudinal or symmetrical axis of a structural member.

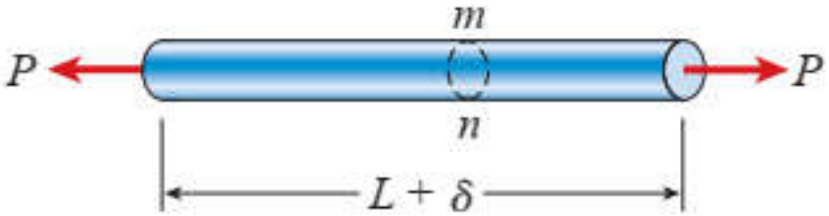


- Internal forces and moments are generated in any section of the body to balance and react to the external loads on a body.



$$Normal\ Stress,\ \sigma = \frac{N}{Area_{mn}}$$

The unit of stress is Pa or N/m^2

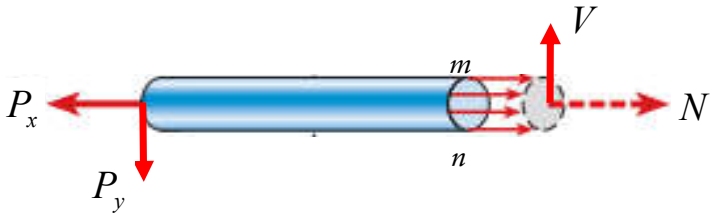


$$Normal\ Strain,\ \varepsilon = \frac{\delta}{L} = \frac{\Delta L}{L}$$

Strain is dimensionless.

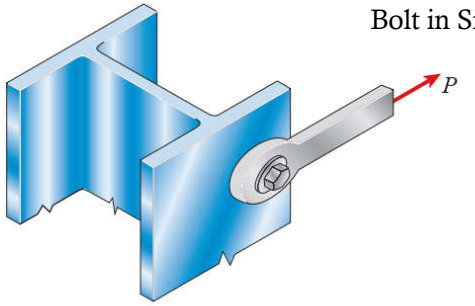
Axial loads; Average Shear Stress

- Axial loads are might be applied on two axes that are parallel to the centroidal axes, or on an inclined plane.

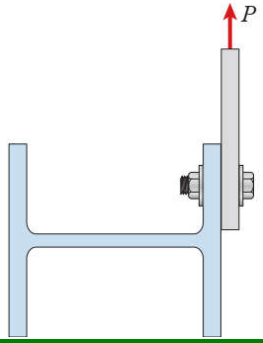


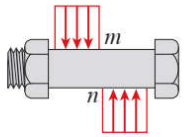
$$Average\ Shear\ Stress,\ \tau_{avg} = \frac{V}{Area_{mn}}$$

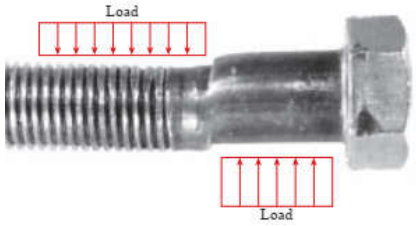
Bolt in Single Shear ($V=P$)

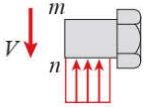


(a)

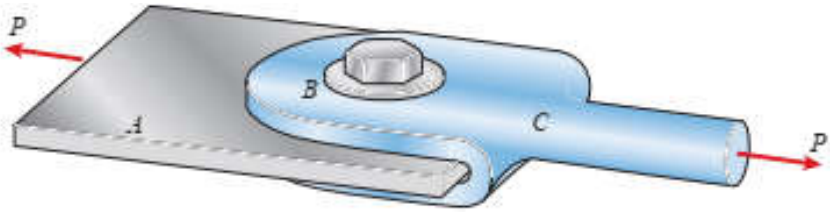




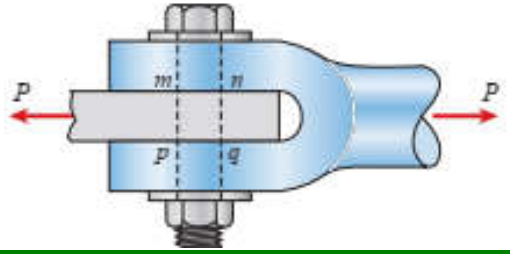


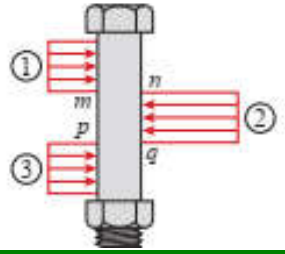


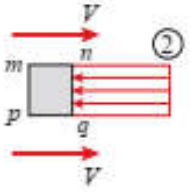
Bolt in Double Shear ($V=P/2$)



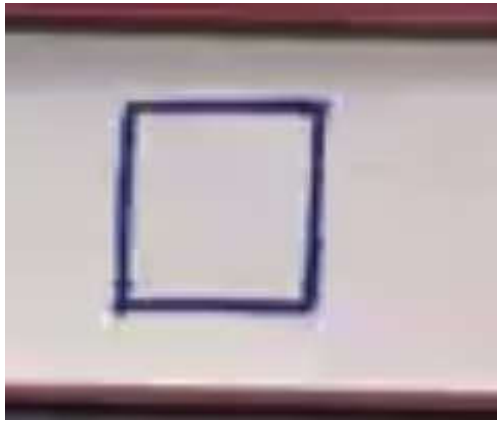
(a)

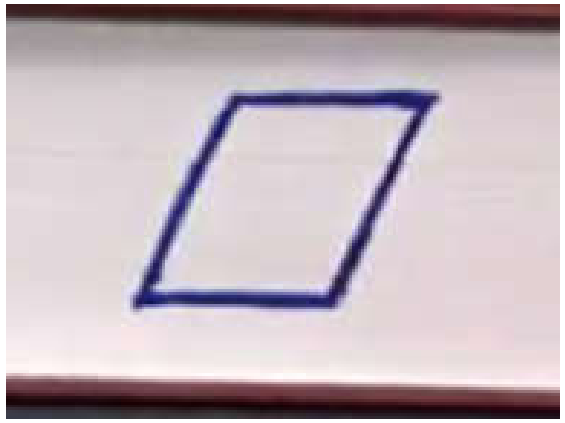


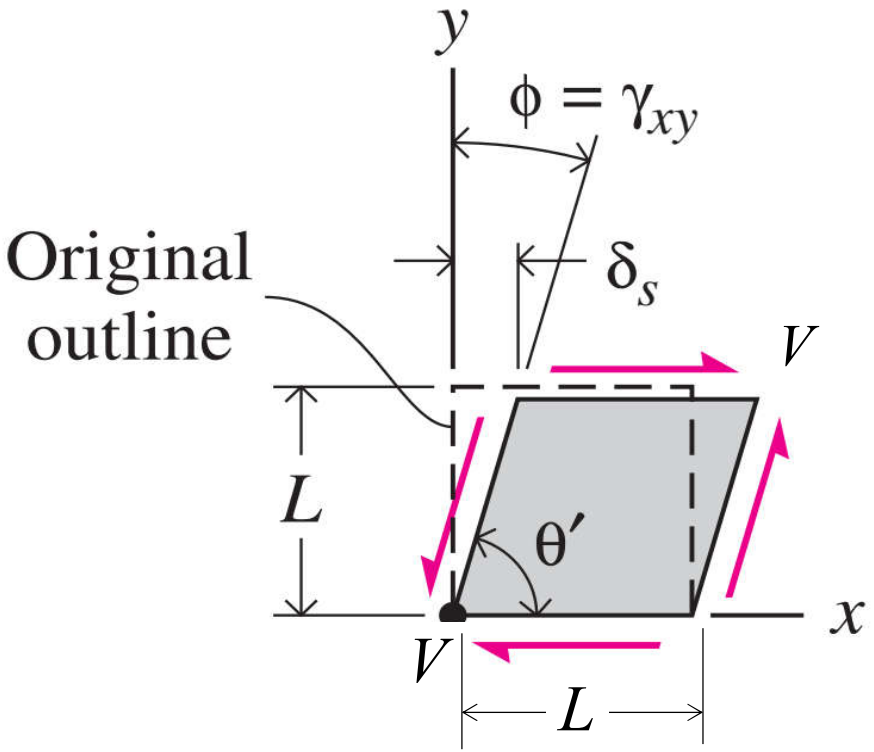




Axial loads; Shear Strain

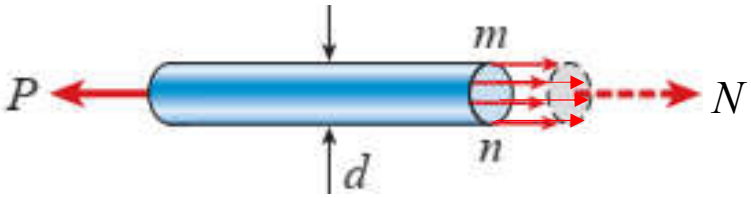




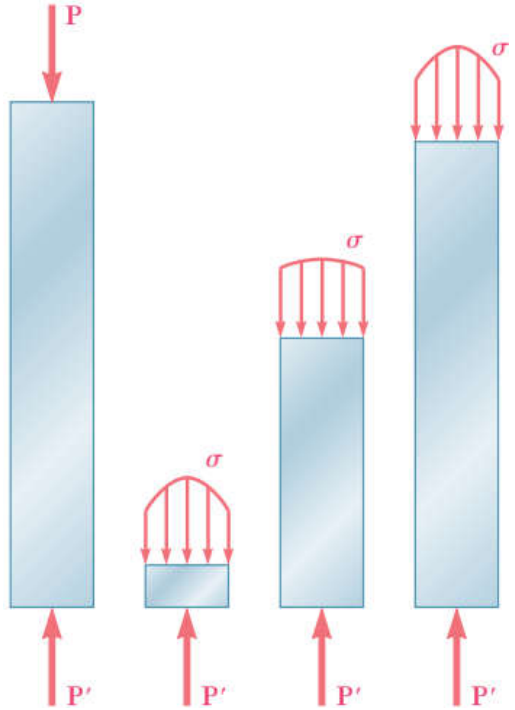


$$Shear\ Strain,\ \gamma = \frac{\delta_s}{L} = \tan\phi = \tan\left(\frac{\pi}{2} - \theta'\right)$$

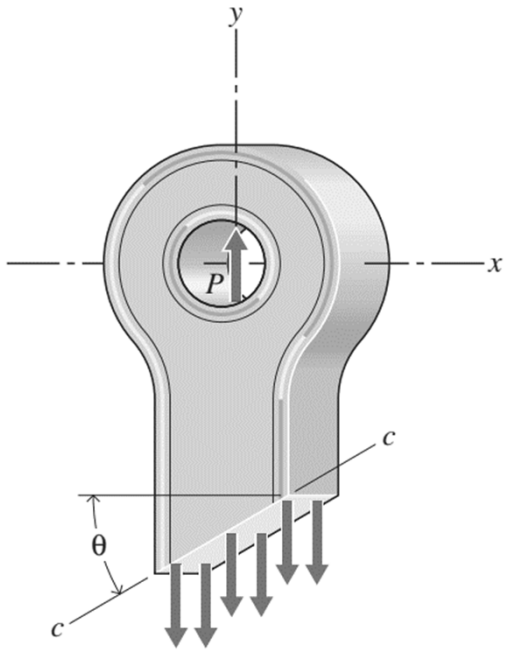
Axial loads; Normal Stress and Strain

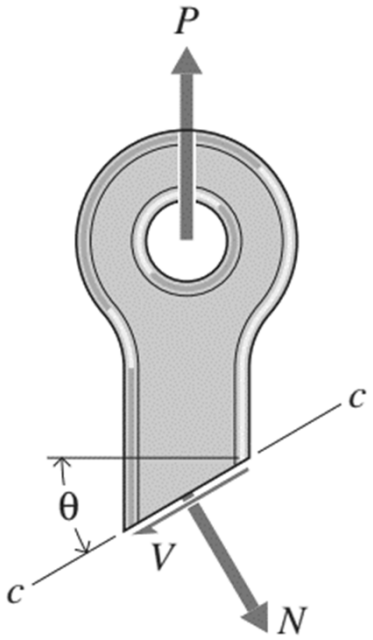


- For the stress to be considered normal, $N \perp \text{Area}_{mn}$.
- For shear, $V \parallel \text{Area}_{mn}$
- Distribution of σ is uniform over the cross-sectional area of the member.
- The stress distribution remains constant over the length of the member, if its cross section remains the same. (*Saint Venant's Principle*)



Axial Loads; Stresses on an inclined plane





$$\textit{Normal Stress, } \sigma = \frac{N}{A_{cc}} = \frac{P\cos\theta}{A/\cos\theta} = \frac{P}{A}\cos^2\theta = \frac{P}{A}(1+\cos\ 2\theta)$$

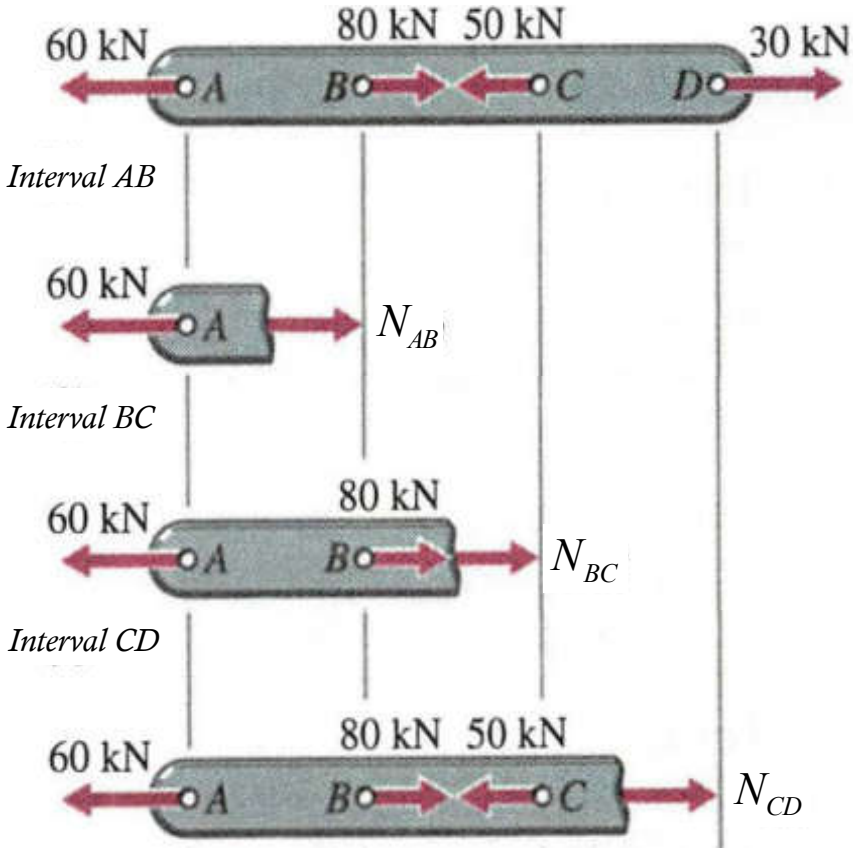
$$\textit{Shear Stress, } \tau = \frac{V}{A_{cc}} = \frac{P\sin\theta}{A/\cos\theta} = \frac{P}{A}\sin\theta\cos\theta = \frac{P}{A}(1+\sin\ 2\theta)$$

Axial loads; Normal Stress

Example

Determine the internal forces transmitted in intervals AB, BC, and CD of the bar shown below. Also determine the stress at each interval if the bar has a constant cross section area of 100 cm^2.





$$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + N_{AB} = 0$$

$$\Rightarrow N_{AB} = 60\text{ kN}$$

$$\sigma_{AB} = \frac{N_{AB}}{A_{AB}} = \frac{60\times 1000\ N}{100\times 10000^{-1}\ m^2} = 6\text{ MPa (T)}$$

$$\overset{+}{\rightarrow}\sum F = 0: \; -60\text{ kN} + 80\text{ kN} + N_{BC} = 0$$

$$\Rightarrow N_{BC} = -20\text{ kN}$$

$$\sigma_{BC} = \frac{N_{BC}}{A_{BC}} = \frac{-20\times 1000\ N}{100\times 10000^{-1}\ m^2} = 2\text{ MPa (C)}$$

$$N_{CD} = 30\text{ kN}$$

$$\sigma_{CD} = \frac{N_{CD}}{A_{CD}} = \frac{30\times 1000\ N}{100\times 10000^{-1}\ m^2} = 3\text{ MPa (T)}$$

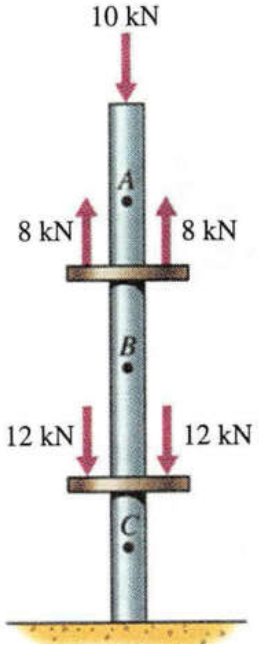
Axial loads; Normal Stress

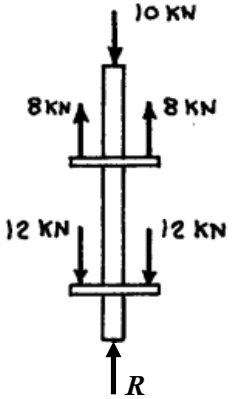
Example:

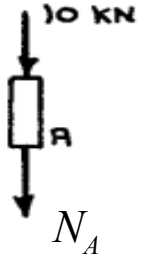
A bar is loaded and supported as shown.

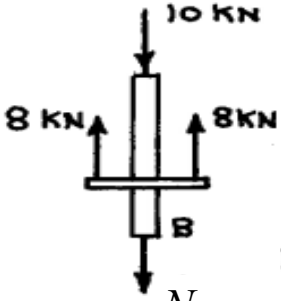
a. Determine the internal forces and stresses at in cross-sections at A, B, and C of the bar. The diameter of the bar is 20 cm

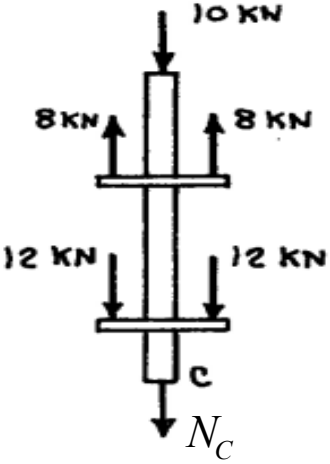
b. Sketch a stress diagram.

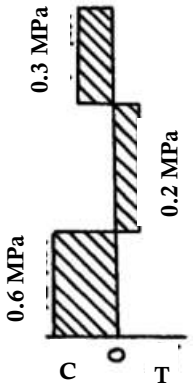












$+\uparrow\sum F=0: R=18\ \text{kN}$

$+\uparrow\sum N_A=-10\ \text{kN}$

$\sigma_A=0.318\ \text{MPa (C)}$

$+\uparrow\sum N_B=6\ \text{kN}$

$\sigma_B=0.191\ \text{MPa (T)}$

$+\uparrow\sum N_C=-18\ \text{kN}$

$\sigma_C=0.573\ \text{MPa (C)}$

Axial loads; Normal Stress

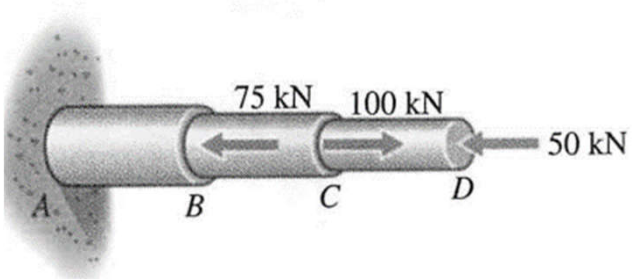
Example:

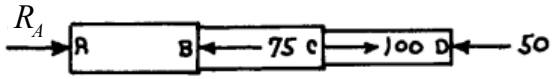
A steel bar has axial loads applied as shown below.

a. Determine the internal forces in intervals AB, BC, and CD of the bar.

b. Determine the normal stresses in each section if the diameters of the intervals are as follows;
AB – 15 cm
BC – 12 cm
CD – 10 cm

c. Sketch a stress diagram.

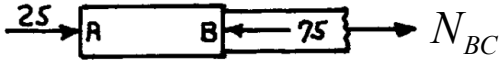




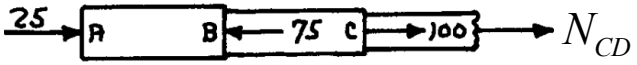
$$\overset{+}{\rightarrow}\sum F = 0: R_A - 75\text{ kN} + 100\text{ kN} - 50\text{ kN} = 0$$
$$\Rightarrow R_A = 25\text{ kN}$$



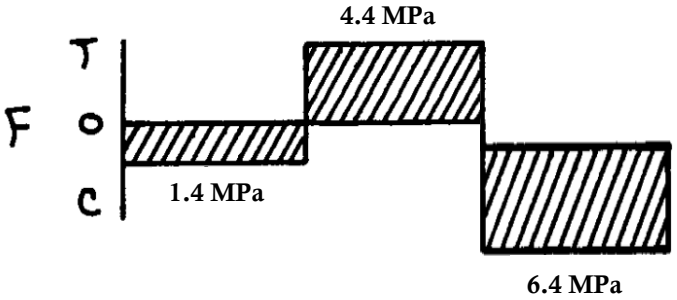
$$N_{AB} = -25\text{ kN},\quad \sigma_{AB} = \frac{N_{AB}}{A_{AB}} = \frac{-25000}{\pi(0.15^2)/4} = 1.415\text{ x}10^6\text{ N/m}^2 = 1.415\text{ MPa (C)}$$



$$N_{BC} = 50\text{ kN},\quad \sigma_{BC} = 4.421\text{ MPa (T)}$$



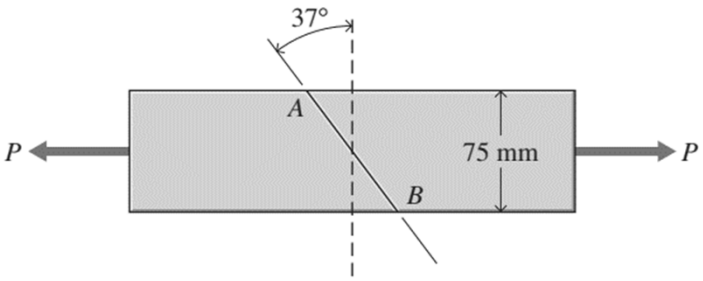
$$N_{CD} = -50\text{ kN},\quad \sigma_{CD} = 6.366\text{ MPa (C)}$$

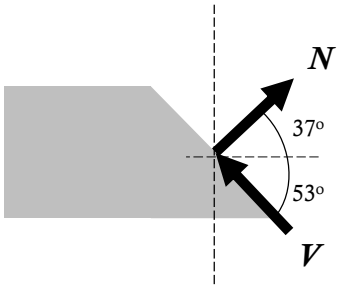


Axial loads; Shear Stress

Example:

The steel bar shown below will be used to carry an axial tensile load of 400 kN. If the thickness of the bar is 45 mm, determine the normal and shearing stresses on plane AB.





$$+\nwarrow \quad \sum F_{\parallel} = 0: \quad V + 400 \times \cos 53^0 = -240.73 \text{ kN}$$

$$+\nearrow \quad \sum F_{\perp} = 0: \quad N - 400 \times \sin 53^0 = 319.45 \text{ kN}$$

$$A_{AB} = 45 \times l_{AB} = 45 \times \frac{75}{\cos 37} = 4225.96 \times 10^{-6} \text{ mm}^2$$

OR

$$A_{AB} = \frac{45 \times 75}{\cos 37} = 4225.96 \times 10^{-6} \text{ mm}^2$$

$$\sigma_N = \frac{319.45 \text{ kN}}{4225.96 \times 10^{-6} \text{ mm}^2} = 75.6 \text{ MPa (T)}$$

$$\tau = \frac{240.73 \text{ kN}}{4225.96 \times 10^{-6} \text{ mm}^2} = 57 \text{ MPa}$$

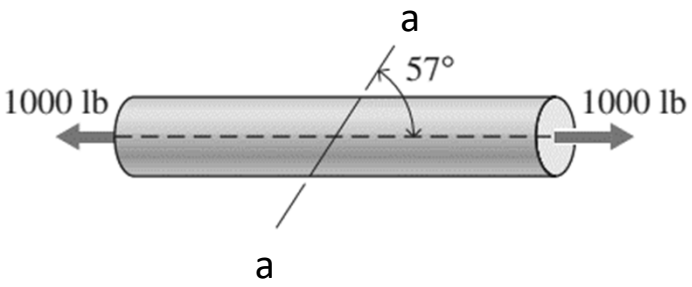
Axial loads; Shear Stress

Example:

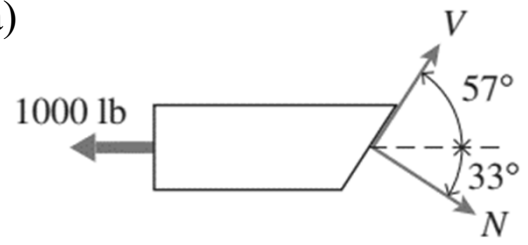
A plastic bar with a circular cross section of diameter 1.25 will be used to support an axial load of 1000 lb as shown below.

a) Determine the normal and shearing stresses on section a-a

b) If the allowable normal and shearing stresses in the adhesive joint used to connect the two sections of the bar are 675 psi and 350 psi, respectively, determine the minimum diameter d for the bar.



a)



$$+\nearrow \quad \sum F_{\parallel} = 0: \quad V = 544.64 \text{ lb}$$

$$+\searrow \quad \sum F_{\perp} = 0: \quad N = 838.67 \text{ lb}$$

$$A = \frac{\pi \times (1.25)^2}{4} = 1.23 \text{ in}^2$$

$$A_{aa} = \frac{A}{\sin 57^0} = 1.46 \text{ in}^2$$

$$\sigma_N = \frac{838.67 \text{ lb}}{1.47 \text{ in}^2} = 573.16 \text{ psi (T)}$$

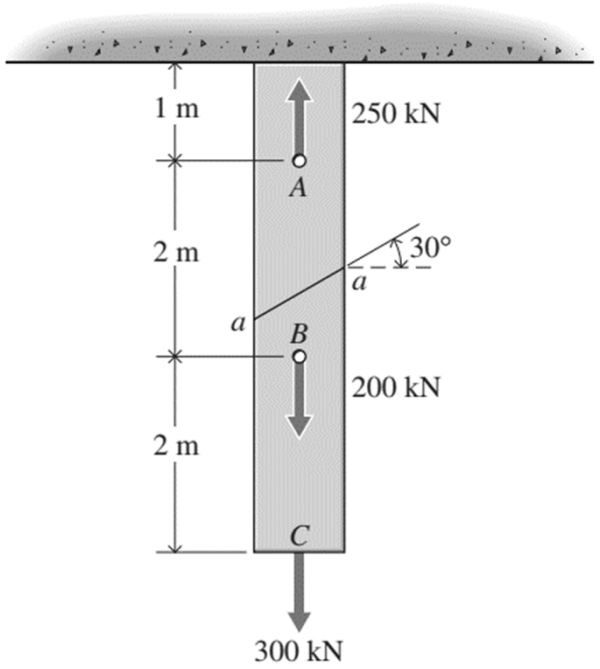
$$\tau = \frac{544.64 \text{ lb}}{1.47 \text{ in}^2} = 372.21 \text{ psi}$$

b) d = 1.289 in

Axial loads; Shear Stress

Example:

The bar shown in Figure shown has a 200 x 100-cm rectangular cross section. Determine the normal and shearing stresses on plane *a–a*.



Normal stress = 18.75 MPa(T)
Shear stress = 10.83 MPa

Axial loads; Normal Stress

Example

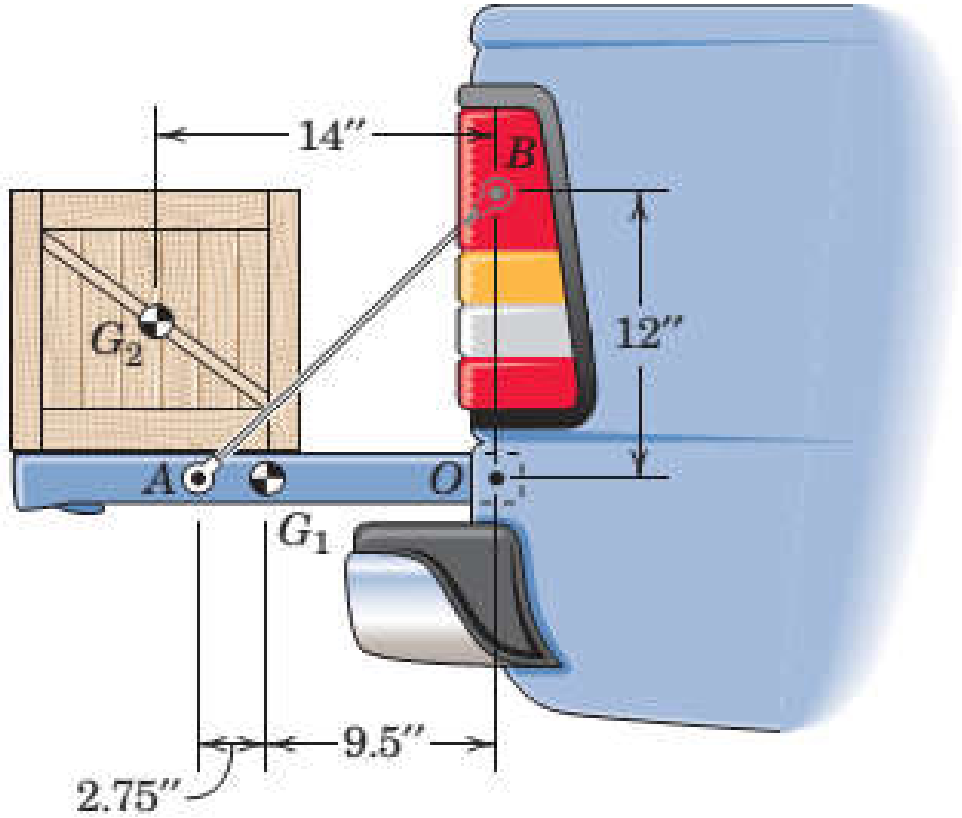
A 150 N crate rests on the 100 N pickup tailgate as shown. The centres of gravity of the gate and crate are at G_1 and G_2 respectively. Assuming the crate is located midway between the two cables;

a) Calculate the tension T in each of the two restraining cables, one of which is shown.

b) If the effective diameter of each cable is 3 cm, determine the normal stress in each cable.

c) If each cable elongates by 0.01 in, determine the average strain in each cable.

d) If the pin at B is in single shear, determine it's diameter if the shear stress on the pin is 2 MPa.



Axial loads; Normal Stress

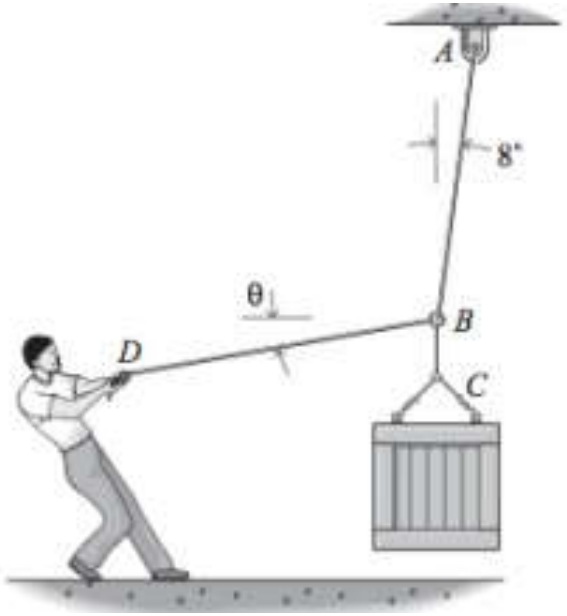
Example:

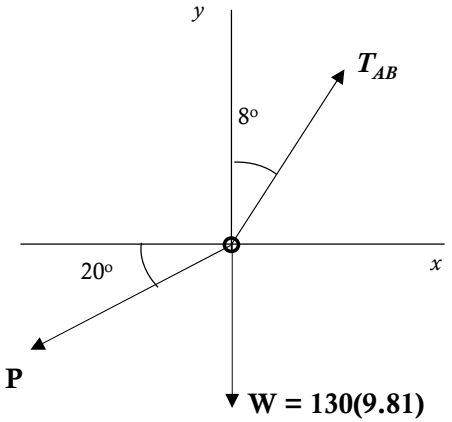
In order to hold a 130-kg crate in a stationary position, a worker exerts a force P at an angle θ on a rope as shown in the figure below.

a) Determine the force in rope AB.

b) Determine the maximum normal stress in the 15 mm diameter rope *AB* when $\theta = 20^0$.

c) Assuming the stress in the rope is rather 5 MPa, what is the diameter of the rope.





$$\pm \rightarrow \sum F_x = 0: T_{AB}\sin 8^o - P\cos 20^o = 0$$

$$+\uparrow \sum F_y = 0: T_{AB}\cos 8^o - P\sin 20^o - 130(9.81) = 0$$

$$T_{AB} = 1357.3\ \text{N}$$

$$\sigma_{AB} = \frac{T_{AB}}{A_{AB}} = \frac{1357.26}{\pi(0.015)^2/4} = 7680250.464\ \text{N/m}^2 = 7.68\ \text{MPa (T)}$$

Axial loads; Shear Stress

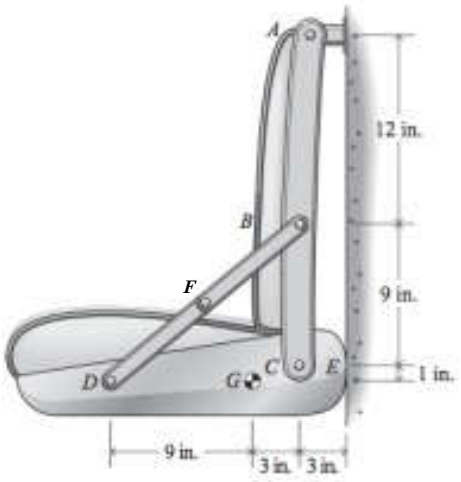
Example:

The fold-down chair shown weighs 30 lb and has its center of gravity at G.

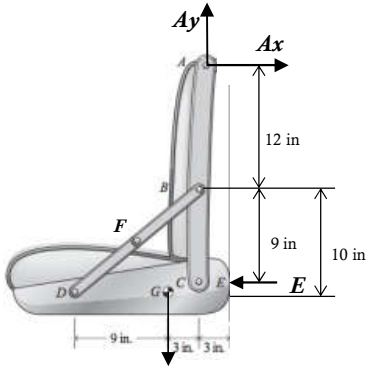
a) Determine the maximum shearing stress on the 3/8-in diameter pin at B, assuming it is in single shear.

b) Also determine the maximum axial stress in link BF if it has an effective rectangular cross-section of dimensions 1" x 0.5"

c) the stress in link DF?



Recall Analysis of Frames from Statics of Rigid Bodies

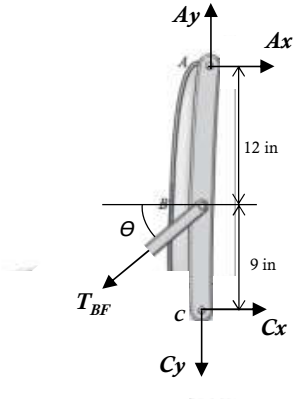


$$\overset{+}{\rightarrow}\sum F_x = 0: \quad A_x - E = 0$$

$$+\uparrow \sum F_y = 0: \quad A_y - 30 \text{ lb} = 0$$

$$\overset{+}{\circlearrowleft}\sum M_A = 0: -3(30) + 21E = 0$$

$$E = 4.2857 \text{ lb} \quad A_x = 4.2857 \text{ lb} \quad A_y = 30 \text{ lb}$$



$$\overset{+}{\circlearrowleft}\sum M_C = 0: \quad 21(A_x) - 9(T_{BF}\cos\theta) = 0$$

$$T_{BF} = 13.0171 \text{ lb}$$

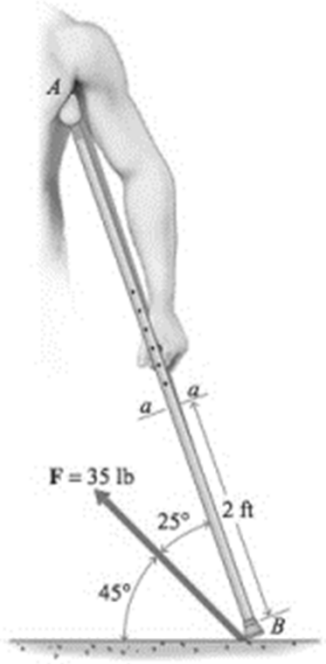
$$\tau_B = \frac{V_B}{A_B} = 117.9 \text{ psi} \qquad \sigma_{BF} = \frac{T_{BF}}{A_{BF}} =$$

Axial loads; Direct and Shear Stress

Example:

The reaction between a crutch and the ground is 35 lb, as shown.

If the crutch is made of a hollow pipe with internal external diameters of 70 mm and 80 mm respectively, determine the normal and shear stresses on plane a-a of the crutch.

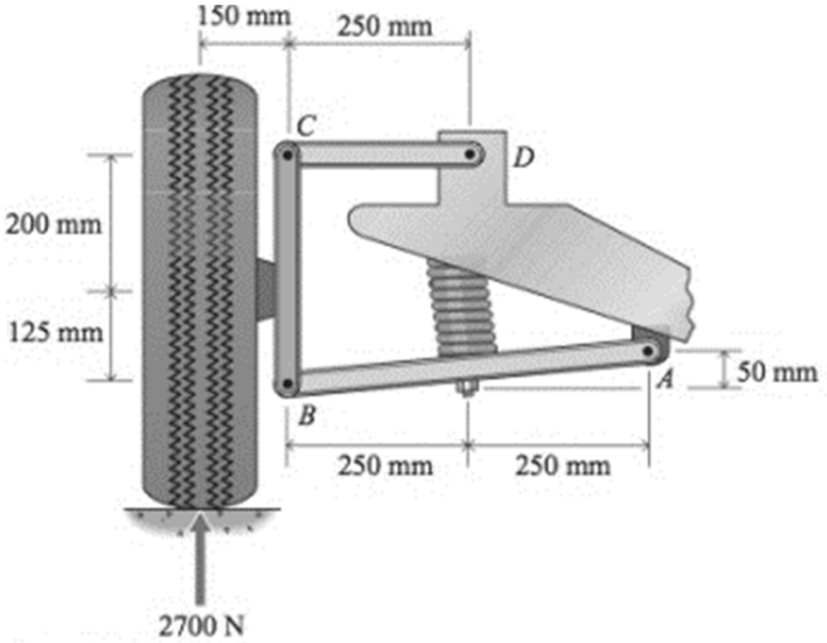


Normal force = 31.7 lb (C)
Shear Force = 14.79
Moment (about the cut) = 29.6 lb.ft (CW)

Example:

The front-wheel suspension of an automobile is shown below. The pavement exerts a vertical force of 2700 N on the tire.

Calculate the stresse(s) on member CD, assuming it is in circular in cross section with a diameter of radius 20 mm.



Normal force = 1246 N(C)
Shear Force = 0
Moment (about the cut) = 0